

Internet Usage and Educational Attainment Around the World

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Agenda

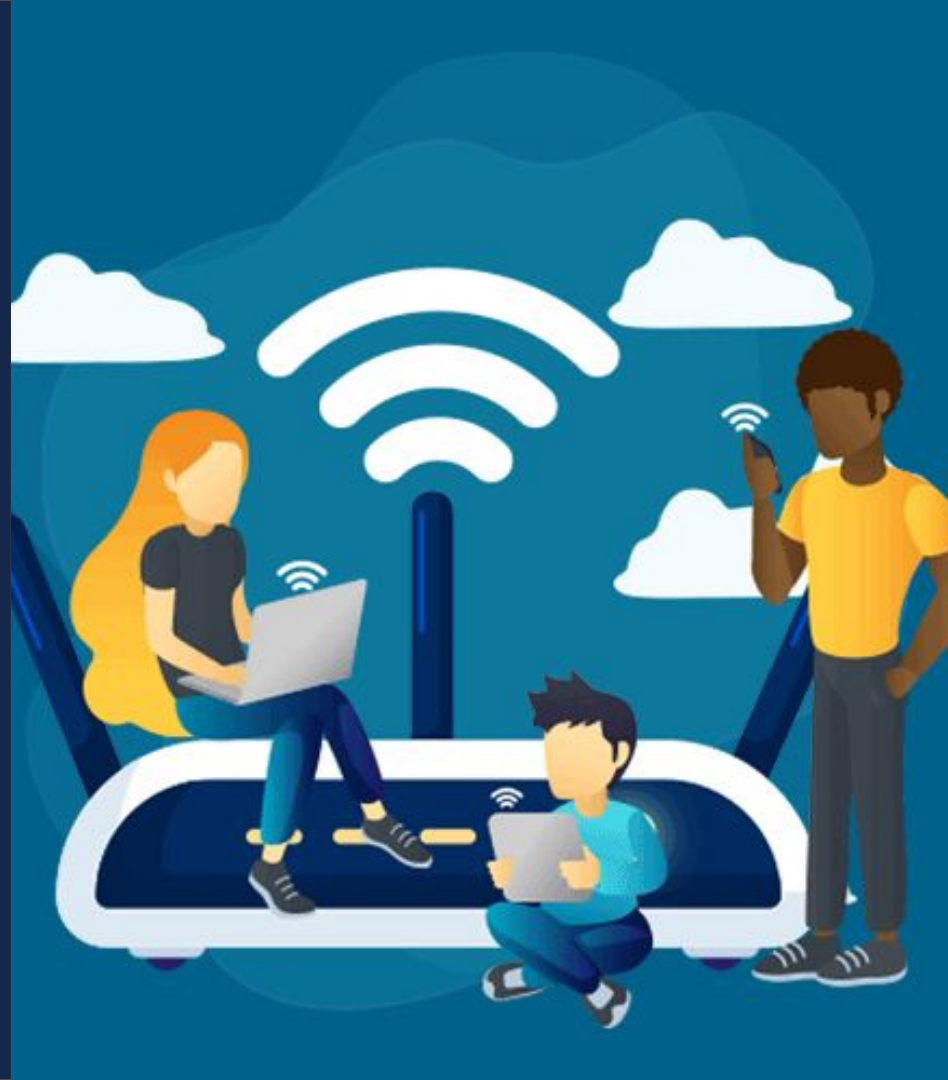
For our reference. No need to keep for actual presentation

- Research Question
- Key Concepts & Variables
- Data
 - Source
 - Wrangling
 - Visualization
- Model
- Assumptions
- Results
- Key takeaways
- Questions

Research Question

*What is the relationship between **internet usage** and **educational attainment** across countries?*

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Data Source

World Development Indicators (WDI) – World Bank

- Cross-sectional country-level data (2022)
- Trusted by IMF, UN, governments
- Indicators used:
 - Internet usage (% of population)
 - Adult literacy (% age 15+)
 - Youth literacy (% age 15–24)
 - GDP per capita (log-transformed)



Key Concepts & Variables

1. Internet Usage (% of Population)

- Measures individuals who have used the internet in the past 3 months
- Proxy for digital access and exposure to online resources

2. Literacy Rates (% Literate)

- **Youth Literacy (ages 15–24):** Basic reading/writing/math skills
- **Adult Literacy (ages 15+):** Broader population measure
- Sourced from UNESCO and national surveys

3. Literacy Difference

- **Youth Literacy – Adult Literacy**
- Captures inter-generational literacy progress

4. GDP per Capita (log-transformed)

- Proxy for national wealth
- Used to control for structural differences in development

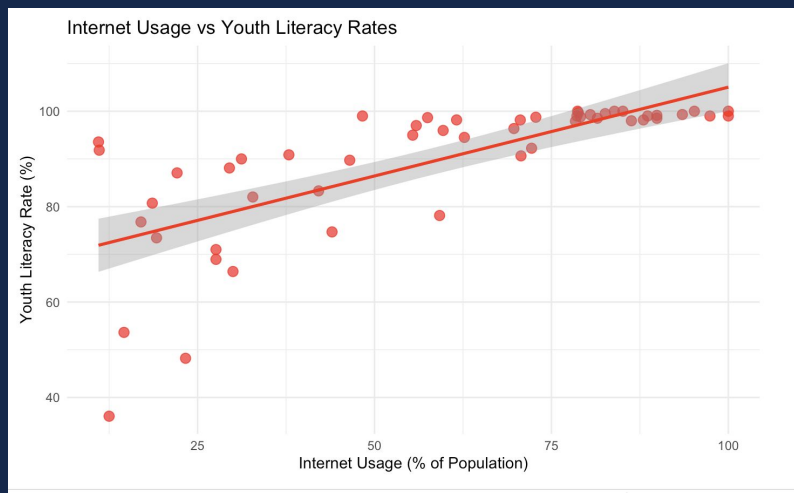


Data Wrangling

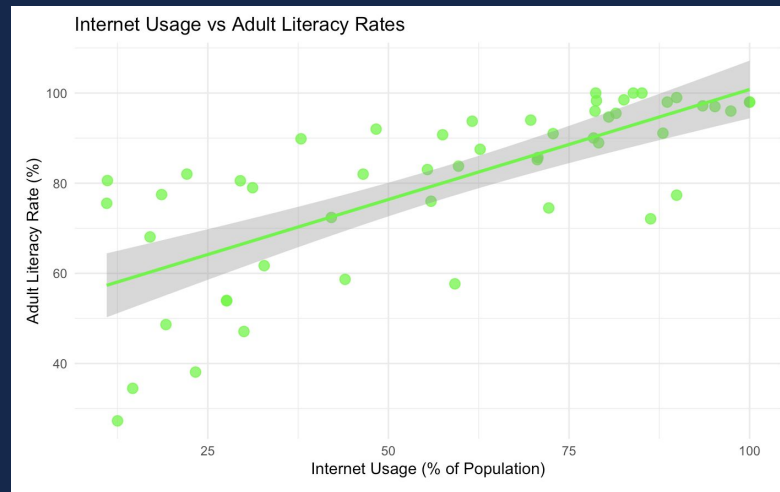
- Filtered complete 2022 data (no missing key values)
- Removed regional aggregates
- Created a new variable:
Literacy Difference = Youth Literacy – Adult Literacy
- Added $\log(\text{GDP per capita})$ to control for national wealth

Data Visualization

Internet Usage vs Literacy Rates

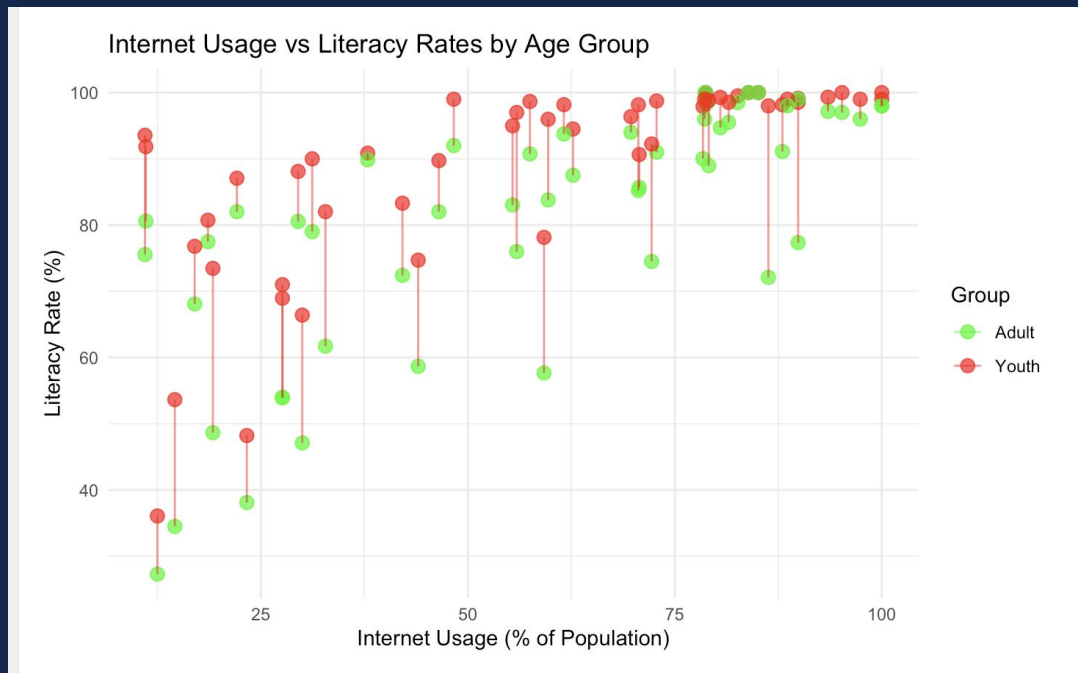


Youth Literacy



Adult Literacy

Data Visualization



Model Specifications

Model	Dependent Variable	Independent Variables
Model 1	Literacy Difference	Internet Usage
Model 2	Adult Literacy Rate	Internet Usage
Model 3	Youth Literacy Rate	Internet Usage
Model 4	Literacy Difference	Internet Usage + $\log(\text{GDP per capita})$
Model 5	Adult Literacy Rate	Internet Usage + $\log(\text{GDP per capita})$
Model 6	Youth Literacy Rate	Internet Usage + $\log(\text{GDP per capita})$

All models are estimated using robust standard errors (HC1)

Model Assumptions

Two Key Assumptions for Large Sample Linear Modeling

1. I.I.D. (independent countries in one year) → **Reasonable**
 - a. Each country is an independent observation
 - b. Drawn from the same data generating process

2. Unique Best Linear Predictor (no perfect collinearity)
 - a. Correlation: 0.877

 - b. VIF: 4.32 → **Acceptable**

Used **robust standard errors** to relax homoscedasticity

Results and Interpretations

Regression Results with Robust Standard Errors

Dependent variable:						
	Lit Gap Model 1 (1)	Adult Lit Model 2 (2)	Youth Lit Model 3 (3)	Lit Gap Model 4 (4)	Adult Lit Model 5 (5)	Youth Lit Model 6 (6)
Internet Usage	-0.12*** (0.03)	0.49*** (0.08)	0.37*** (0.07)	0.03 (0.08)	0.37** (0.16)	0.40*** (0.12)
log(GDP per Capita)				-3.62** (1.61)	2.95 (2.82)	-0.67 (2.09)
Constant	15.83*** (2.07)	51.98*** (5.86)	67.81*** (5.34)	36.72*** (8.92)	34.94** (15.63)	71.66*** (11.47)
Observations	52	52	52	52	52	52
R2	0.20	0.54	0.52	0.29	0.55	0.52
Adjusted R2	0.18	0.53	0.51	0.26	0.53	0.50

Note:

*p<0.1; **p<0.05; ***p<0.01

Results and Interpretations

Analysis of Variance Table

Model 1: Literacy_Difference ~ IT.NET.USER.ZS

Model 2: Literacy_Difference ~ IT.NET.USER.ZS + log(NY.GDP.PCAP.CD)

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	50	2160.8				
2	49	1919.1	1	241.64	6.1696	0.01646 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Analysis of Variance Table

Model 1: SE.ADT.LITR.ZS ~ IT.NET.USER.ZS

Model 2: SE.ADT.LITR.ZS ~ IT.NET.USER.ZS + log(NY.GDP.PCAP.CD)

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	50	8189.2				
2	49	8028.5	1	160.69	0.9807	0.3269

Analysis of Variance Table

Model 1: SE.ADT.1524.LT.ZS ~ IT.NET.USER.ZS

Model 2: SE.ADT.1524.LT.ZS ~ IT.NET.USER.ZS + log(NY.GDP.PCAP.CD)

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	50	5057.5				
2	49	5049.2	1	8.2275	0.0798	0.7787

Key Takeaway

LITERACY

Positive correlation between internet usage and literacy rates across countries

1

LITERACY GAP

Youth-adult literacy gap narrow as internet access increases but this effect disappears once GDP is included

2

SO WHAT...

Access to the internet is a power tool for advancing education

3

WHAT ELSE?

Effect of internet usage on education goes beyond literacy (ex: higher education, research & development)

4

Thank you for Listening!

Q&A

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